
What Does a Protein Surface Add? Ablations Cannot Tell You

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Abstract

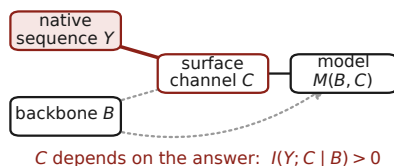
When a model’s input is *computed* rather than given, the ablation that removes it cannot say why it helped: the channel may add information derived from the prediction target (Route A), or merely re-represent information the model already has (Route B), and both produce the same gap. No precision separates them — that is a property of the estimand, not of the sample. We give a seven-component provenance-aware protocol that does separate them, and run it on released surface-conditioned inverse folding. In two released pipelines the surface chemistry channel is a deterministic function of native residue identity: 89.12% and 90.12% of its values move under fixed-coordinate permutation, against 0.00% for geometry. For the system with released weights, a fixed code-table derangement makes it write the counterfactual protein, raising agreement with that sequence by +38.90 pp over its own pre-intervention baseline — a targeted error pattern that generic distribution shift does not explain. Across paired retraining seeds, a semantics-free random 18-class partition gains far more than scrambling biochemical values over the same classes: the benefit follows target-identifying partition structure, not mainly biochemical meaning. At a matched $\approx 17\%$ marginal ceiling, target-derived channels add +12.49 pp to frozen ProteinMPNN and +11.07 pp to frozen ESM-IF1 while backbone-derived channels are ignored — holding unfitted on an external RCSB 2023+ panel across three frozen families. Joint surface instruments remain below our Route-B threshold, while a backbone-derived positive control clears it. Four of our own predeclared predictions were falsified and a fifth claim family was retracted; all five are retained. The claim is a provenance burden: whenever a method computes an input from data involving the prediction target, an ordinary ablation cannot justify a representation claim until source interventions separate added target information from design-valid computation.

1 Introduction

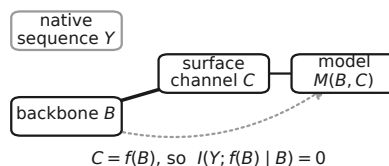
Surface-vs-no-surface ablations are underidentified: when a method computes an input from data it may see, removing that input cannot tell whether the gain came from target-derived information or legitimate re-representation. This is an identification failure, not a protein-specific quirk (§1.3). Derived *labels* and contaminated *splits* are policed; derived preprocessing *inputs* receive less scrutiny. We study the failure in surface-conditioned protein inverse folding.

Inverse folding predicts sequences for a given backbone (Kuhlman et al., 2003; Ingraham et al., 2019; Dauparas et al., 2022; Hsu et al., 2022), and became broadly applicable once accurate structure prediction made novel backbones cheap (Jumper et al., 2021). Methods increasingly condition on a molecular surface too (Song et al., 2024; Wu et al., 2026), reporting gains via surface-vs-no-surface ablations. But a surface is *computed from structure*, and while some construction choices depend

Route A — information addition



Route B — design-valid representation



Both structures produce the same ablation gap $\Delta = M(B, C) - M(B, \emptyset)$. The gap does not identify which one is operating.

Figure 1: Two causal structures, one ablation gap. In Route A the surface channel is computed from the native sequence, so it carries information the backbone does not; in Route B it is a deterministic function of the backbone, so it carries none. Both produce the same surface-versus-no-surface difference, which is why that difference does not identify which one is operating.

only on the backbone, others depend on the native sequence — the prediction target. The ablation sums both routes (Figure 1).

1.1 Two routes

Route A — information addition. The surface construction depends on the native sequence Y , so the channel carries information about Y the backbone B does not contain, and the model can improve by reading it. Under the inverse-folding contract that information is unavailable at design time: for a novel backbone, Y is unknown by definition.

Route B — design-valid representation. The surface is a deterministic function $C = f(B)$ of the backbone alone, so

$$I(Y; f(B) | B) = 0$$

by construction — the channel adds no information. A jointly trained model may still benefit, but only through representation, optimisation or finite-capacity extraction. The Route-B identity is the motivation for our contribution: a protocol that determines, for a real pipeline, which route operates and what each is worth.

1.2 What we contribute

1. A seven-component identification protocol (§3.1). Its load-bearing assembly establishes target dependence, holds capacity fixed while varying provenance, and admits representation gains frozen models cannot express; dropping any of components {1,3,6} underidentifies the ablation again.
2. A released-checkpoint mechanism result (§4): a fixed code-table derangement makes the model transcribe the counterfactual sequence. Across two released pipelines, target-derived chemistry moves with the native sequence; destroying biochemical semantics preserves nearly all of its benefit; and provenance-matched frozen and joint controls separate Route A from Route B.
3. A general provenance burden with three cheap checks (§5): intervene on source, bound target capacity, and compare equally decodable target-derived and design-valid channels. The resulting contribution is a diagnostic protocol and evidence standard.

1.3 Why an ablation cannot separate the routes

Let a surface-conditioned model receive backbone B and surface channel C , and let $M(B, C)$ denote recovery. The usual ablation reports

$$\Delta_{\text{ablation}} = M(B, C) - M(B, \emptyset)$$

This estimand is compatible with either route or any mixture. Two pipelines with identical Δ_{ablation} can differ completely: one with C independent of Y given B (pure Route B), the other with C a near-perfect function of Y (pure Route A). No precision separates them, because the estimand does not distinguish provenance — Appendix F builds both explicitly, matches their gaps to 0.07 pp across five seeds, and separates them with one intervention. Separating the routes needs interventions holding one fixed while varying the other, which is what §3.1 constructs.

2 Related work

SurfDesign is the closest substrate: its surface ablation does not separate target-derived chemistry from a backbone-only surface control, which §§4.3–§4.4 provide. Wu et al. (2025) remove a related side-chain-to-sequence shortcut by redesigning a generator; we instead identify provenance in released pipelines. Surface-fingerprint interaction design (Gainza et al., 2023) and refolding-evaluator limitations (Korbeld et al., 2026) establish that surfaces can be useful and evaluators unreliable, but neither resolves provenance. None of these works establishes the underidentified estimand or combines our source, capacity, matched-decodability, transfer and mechanism checks. Other systems also differ: one construction is backbone-derived (Bai et al., 2025), whereas several surface pipelines use residue identity. We make no prevalence claim; this variation motivates a protocol, not a family-wide generalisation.

3 Methods

3.1 The identification protocol

Table 1 sets out the protocol.

Table 1: The identification protocol. Components 1–3 are input-side and need no trained model, 4–6 are model-side, and 7 opens the trained model.

#	Test	Identifies
1	Fixed-coordinate substitution	target dependence
2	Channel-only Bayes ceiling	available target information
3	Matched marginal ceiling	provenance at fixed capacity
4	Native / predicted / constant / shuffled	alignment versus prior
5	Frozen published models	incremental information utility
6	Matched joint training	design-valid representation
7	Remove / restore / repair pathway	commitment location and cost

Only {1, 3, 6} separate the routes; {4, 5, 7} characterise how the channel is used, its strength at published scale, and its mechanism. The *assembly* is what identifies, because each alone is insufficient: substitution shows dependence on Y but not that the ablation gap is carried by it, a capacity bound is compatible with a design-valid geometric channel, and joint training reproduces an underidentified ablation if the surface still reads the target. Every load-bearing conclusion needs a *pair*, which is why this is a protocol rather than a survey of known checks.

3.2 Systems, populations, and accounting

Subject pipeline. We audit the released SurfDesign input pipeline at commit 9ce3039 (Wu et al., 2026), with its preprocessing, configuration and E(n)-GNN encoder source unchanged (Satorras et al., 2021). Because no published checkpoint exists, trained attribution results use that encoder under matched data and budgets with our controlled decoder; they are not a numerical reproduction of the full published system. For orientation against that system, its paper reports 74.13% median recovery on CATH 4.2 and, as leakage-oriented controls, a backbone-only variant at 65.12% amino-acid recovery (AAR) and an ESMFold-predicted-structure setting at 68.5% AAR. It also answers the provenance question in words: its §2.1 states that the surface features are “computed from local atomic geometry and element types, not from residue identity labels”, and that the pipeline does “not

use: (i) residue identities” during surface generation. Its own appendix names hydrophobicity and charge as “pivot to the performance improvement” — the two that read residue identity (Appendix D). The released `computeHydrophobicity` is a twenty-entry Kyte–Doolittle table indexed by three-letter residue name, and §4.1 measures 89.12% of its output moving when identities permute at fixed coordinates. We report the gap between a stated scope and a released implementation and make no claim about how it arose. The two controls are context, never contrasts we compute against: the first is the underidentified ablation of §1.3, the second varies *coordinate accuracy* while the chemistry channel still reads the native sequence. They do fix scale — a few points, not the ≈ 50 pp spread our retrain shows (Appendix A). Our trained attribution arms contain no protein language model.

Bridges and controlled forms. Frozen ProteinMPNN v_48_020 (1.66M) and ESM-IF1 esm_if1_gvp4_t16_142M_UR50 (141.66M) — published weights, never trained by us, spanning an $85.3\times$ parameter range with no shared architectural family — plus the official PiFold/ProDesign `Training&Data/checkpoint.pth` (ICLR 2023) in §4.3. The joint Route-B instruments are deliberately small: the released architecture retrained, and the unrelated Simple2 at $\approx 22\%$ backbone-only recovery (Appendix A).

Populations. Family-disjoint CATH 4.2 splits from the graph-based inverse-folding benchmark (Orengo et al., 1997; Ingraham et al., 2019), hash-pinned and used as distributed, under one selection rule (40–260 residues, canonical only, complete backbones), bounding every result to chains of at most 258 residues. Five populations carry the claims — P-RELEASED (3,946/790 chains), P-FROZEN (593/197), P-FROZEN-FULL (928 test chains / 125,401 residues), P-CEIL (195,631 residues, analytic bounds only) and P-TS50/P-TS500 (50/500; O’Connell et al., 2018) — and are never merged: joint-training and frozen-model claims rest on different populations (Appendix A, Table 3).

Experimental scope. §4.1 uses P-CEIL for analytic bounds and a 38-chain substitution panel; §4.2 uses SurfPro’s 887-chain official test split for its released checkpoint and three paired 4,000-update retraining seeds; §4.3 uses P-FROZEN-FULL for frozen ProteinMPNN/ESM-IF1 and the RCSB panel for three frozen families; §4.4 uses P-FROZEN, three paired 3,000-step seeds on two encoders, and 10 seeds for its positive control. Frozen-model intervals resample chains; trained-arm intervals resample seeds and chains. Every readout is native sequence recovery except §4.2’s explicitly paired official and coverage scorers.

The evaluation spans CATH, TS50 and TS500 under one primary data authority, together with an external input-side and matched panel. Overall, the study has 3 named benchmarks, 1 primary data authority, and 0 independent external-data replications. TS50 and TS500 overlap CATH, so they are benchmark transfers, never independent-data generalisation. After exact-overlap removal, CATH+TS contains 1,455 chains and 259,730 residues — scale, not generality. Base rates differ from published ones through population and decoding rule, not reimplementations (Appendix A); all quantities are within-population contrasts.

3.3 Statistical protocol

Uncertainty is matched to each estimand. Every trained contrast has two variance sources — chain sampling and seed draw — and reporting only the first understates the interval. Jointly trained arms therefore use a two-level bootstrap (Efron, 1979) over seeds *and* chains, and that quantity — the contrast in the chain population — carries every headline claim. Frozen-model contrasts, having no training seed, use a paired per-chain bootstrap. §4.4 also reports a descriptive cross-check over seed means at a fixed chain set; it describes the *seed mean*, carries no coverage or tail claim — three seeds cannot calibrate one — and is never compared with a two-level interval. Arms are compared on the intersection of chains every seed scored.

Predeclaration. Each decisive threshold was fixed before the run testing it: +1 pp for a practically meaningful route-B benefit, 4 pp mean absolute error (MAE) for the independence-law test, 1.5 pp for the graded-payoff tests, and directional predictions for each §5 falsifier. We report the predeclared verdict whether or not it favours our framing. The four Route-B contrasts are one predeclared family, reported rowwise without a simultaneous-coverage claim: the two ten-seed rows use two-level 95% intervals, and the two three-seed rows are directional only.

Fitting and reporting. λ is fitted on train only. Micro and macro recoveries are never mixed within a contrast, and the metric throughout is native sequence recovery, not fold quality or function. Optimiser

155 details and draw counts: Appendix A; per-seed constituents: Appendix B. The §3.2 crosswalk exposes
156 the update and seed budgets for every trained headline family.

157 4 Results

158 4.1 Provenance and capacity

159 Provenance (component 1). In the released path the surface hydropathy channel *hp* maps each surface
160 point to its nearest residue and looks that residue up on the Kyte–Doolittle scale (Kyte & Doolittle,
161 1982). Permuting residue identities with coordinates held fixed moves 89.12% of *hp* values across 38
162 chains, while the geometry channel *hbond* moves 0.00%: the chemistry channel is a deterministic
163 function of the native sequence, the geometry channel is not. It falls short of 100% only where a
164 permutation substitutes a residue of equal hydropathy; every position whose value *can* change does.

165 Capacity (component 2). Treating the channel as the only input, Bayes-optimal recovery is 85.89%
166 on 195,631 test residues against an 8.86% frequency floor: the 20 residues map to 17 channel values,
167 78.65% are uniquely resolved, and the sole collision is {D, E, N, Q} at −3.5 (Appendix E). That ceiling
168 is not a property of the chemistry: of the 741,285 possible 17-class tables, 82% have a higher one
169 and the median beats this one by 2.61 pp, so the capacity is structural in partitioning identity at all
170 (Appendix D). Surface points reach 99.99% of residues and 99.94% of buried ones (Table 6 decodes
171 one chain by hand). This is an input-side bound — what the channel *could* supply — which §4.2
172 turns into a decoder and runs.

173 4.2 A released checkpoint transcribes its channel

174 Everything above tests a pipeline with no published weights. SurfPro (Song et al., 2024), an
175 independent ICML surface-conditioned inverse folder, *does* release a CATH 4.2 checkpoint, so the
176 protocol can run on a system as shipped. With its official repository, hash-verified checkpoint and
177 data, its own 887-chain split and scorer, changing only CUDA device placement, we reproduce its
178 reported 57.78% recovery at 57.79%.

179 Its surface files name the nearest residue at every vertex and map that name through a 20-row
180 hydropathy/charge table resolving 20 residues into 18 classes. Component 1 returns §4.1’s verdict on
181 a second implementation: over 3,076,319 vertices with coordinates byte-identical, 90.12% of channel
182 values move under residue permutation and 0.00% of geometry does. Component 2 stops being
183 analytic: an 18-entry per-class argmax with no model, no coordinates and no learned weights, fitted
184 on train only, scores 69.72% [68.93, 70.49] against the released model’s 70.33% [69.53, 71.10] on a
185 coverage metric applied to both — +0.61 pp [−0.05, +1.28] — and 54.67% against 57.79% on the
186 system’s own official scorer, +3.12 pp [+2.46, +3.78]. We report both. A parameter-free table does
187 not equal the model; it lands within a few points of one, which bounds how much of the benchmark
188 score the channel alone can carry. Component 4, at published scale, weights fixed: zeroing the
189 channel costs −54.27 pp [−55.16, −53.37], an in-support constant −52.96 pp, and shuffling values
190 within a chain — same values, same marginal, wrong vertices — −46.69 pp [−47.57, −45.78].

191 The derangement demonstrates transcription rather than generic collapse (Figure 2a). We permute the
192 code table by a derangement π fixed before the run, a fixed-coordinate substitution $a \mapsto \pi(a)$. Native
193 recovery falls from 57.79% to 18.36%, while agreement with the particular counterfactual $\pi(Y)$ rises
194 from 17.75% to 56.65%: +38.90 pp [+38.03, +39.76] and 98.0% of the model’s published-scale
195 recovery. A second prespecified derangement gives 52.45% and a +34.87 pp [+34.05, +35.68]
196 targeted gain over 17.59%. Generic out-of-distribution degradation cannot produce this specific
197 redirection.

198 Its error fingerprint agrees: the model’s four largest confusions are exactly the four ordered pairs
199 inside the table’s only two collision classes (Appendix D). Claim ceiling: components 1, 2, 4 and 5
200 on a released checkpoint, not a route separation for SurfPro, which needs component 6, and nothing
201 about that system’s binder or enzyme tasks, whose contract differs. The subject pipeline’s own
202 retrained arms reproduce the alignment signature at matched budget ($A_2 - A_3 = +0.32$ pp [+0.12,
203 +0.56]; Appendix A).

204 The channel’s value lies primarily in its partition rather than its chemistry. Retraining SurfPro’s own
205 pipeline at matched budget with the code table replaced — three paired optimization seeds, 887

chains per arm and seed — separates the two readings. Scrambling the biochemical values across the same 18 classes moves recovery by at most +0.342 pp, 0.715% of the 47.82 pp the channel is worth against its zeroed control, while a random 18-class partition carrying no chemistry *beats* the released table by +3.347 pp [+2.393, +4.301]. This is not “semantics do not matter”: *scram* - *native* is negative at all three seeds with a paired interval excluding zero, so the specific assignment may matter at the few-tenths level — it just cannot explain the 47.82 pp. Per-seed values, both interval treatments, the 11-arm family and two more failed predictions: Appendix B.

4.3 Route decomposition on frozen published models

We construct two instruments — a random recoding of residue identity and a quantisation of backbone geometry — that hold marginal Bayes decodability fixed while contrasting target- and backbone-derived *sources*. Two seeded target recodings tied exactly on ceiling; choosing between them costs 0.89 pp, and both exclude zero widely (Appendix A). Component 3 equates marginal decodability, not conditional mutual information. Incremental conditional utility is tested separately by the frozen and jointly trained arms, so the conclusion is a provenance-conditioned payoff contrast, not an equality-of-information theorem. On CATH, target- and backbone-derived Bayes ceilings differ by 0.13 pp; the CATH-fit external pairs differ by 0.36–0.84 pp, all remaining near 17% and small beside the 10.80–12.29 pp target-derived gains (Table 2). We inject by a one-parameter rule ($\text{logits} + \lambda \cdot \log P(y \mid \text{code})$, λ fitted on train). They match on marginal ceiling but not on cardinality (two classes against sixteen), a possible confound *in design*; in the outcome the λ grid returns $\lambda = 0$ for the matched backbone channel on all three frozen families and every nonzero backbone coefficient elsewhere in the 48-pair sweep still produced a negative gain, so if extra classes were buying the target-side payoff, backbone codes with *more* classes would not be ignored or hurt recovery (Appendix A).

These are matched *instruments*: the payoff below is that of a $\approx 17\%$ matched ceiling, not of the released 85.89%-ceiling channel. That ceiling is the top of what we could build — the best held-out backbone-derived channel reaches 18.60% — so the 85.89% contrast is un-run for want of a construction, not declined, and a better design-valid channel would raise it (Appendix D).

The result is Table 2, which also carries the external panel.

Table 2: Provenance contrast at approximately matched marginal decodability, in percentage points of native sequence recovery. T/B gives the CATH-fit target/backbone ceilings used for each row; the mismatch is 0.13 pp on CATH and 0.36–0.84 pp after unrefitted external transfer. Train-selected $\lambda = 0$ gives the backbone-derived $\Delta = +0.00$. CATH uses 928 chains / 125,401 residues; RCSB 2023+ uses 68 chains / 9,540 residues for ProteinMPNN and a shared 23 chains / 3,230 residues for ESM-IF1 and PiFold. Paired per-chain bootstrap; cell denominators: Appendix A.

Model	Population (CATH-fit ceiling T/B, %)	target-derived Δ	matched backbone-derived Δ
ProteinMPNN	CATH (17.13/17.00)	+12.49 [+12.29, +12.68]	+0.00
ESM-IF1	CATH (17.13/17.00)	+11.07 [+10.88, +11.27]	+0.00
ProteinMPNN	RCSB 2023+ (17.13/16.76)	+12.00 [+11.32, +12.71]	+0.00
ESM-IF1	RCSB 2023+ (17.13/16.51)	+10.80 [+9.76, +11.90]	+0.00
PiFold	RCSB 2023+ (17.40/16.56)	+12.29 [+11.11, +13.59]	+0.00

The trained nonlinear “combiner”, which *can* go negative, gives -0.03 pp on ProteinMPNN and -0.41 pp on ESM-IF1. Across 48 channel-model pairs the families separate with no overlap — target-derived +9.75 to +48.30 pp, backbone-derived -0.92 to $+0.00$ pp — and not only at the matched pair:

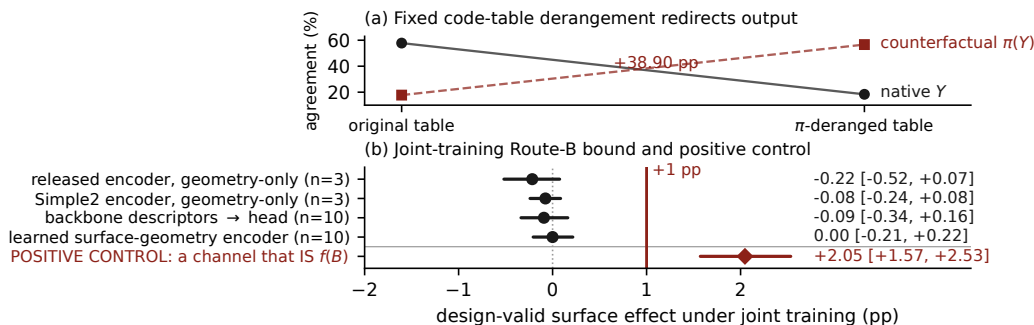


Figure 2: Two interventions separate the routes. (a) On SurfPro’s official 887-chain split, a code-table derangement fixed before the run redirects output from the native sequence toward its specified counterfactual; bars are paired chain-bootstrap 95% intervals. (b) All four qualifying jointly trained point estimates fall left of the predeclared +1 pp threshold. The same apparatus’s positive control clears +1 pp on its lower bound. Two rows use 10-seed marginal 95% intervals; the two three-seed rows are directional checks without coverage claims. Tested conditions and controls: Section 4.4.

236 read the target family’s own capacity–payoff relation at each backbone channel’s ceiling and every
 237 in-range design-valid channel falls at least +9.65 pp short (worst bootstrap 2.5th percentile over four
 238 predeclared estimators, both models), against a grouped-holdout error of 2.26–3.33 pp within the
 239 fitted family — a descriptive interpolation, not a law (Appendix D).

240 The strongest design-valid replacement is a *learned* residue predictor over backbone-only geometry
 241 (19 features) reaches 21.43% standalone, well above P-FROZEN-FULL’s 8.94% floor, yet adds +0.00
 242 pp to both frozen models. That zero is a fitted $\lambda = 0$, so we ran the trained combiner directly on it:
 243 −0.23 pp [−0.36, −0.09] on ProteinMPNN and −0.73 pp [−0.87, −0.58] on ESM-IF1, both excluding
 244 zero. A combiner free to exploit any interaction the additive rule cannot express recovers no benefit
 245 from the strongest design-valid channel we could build.

246 4.4 Route decomposition under joint training

247 A frozen bridge cannot express Route B: fitted without ever seeing the representation, it tests only
 248 whether a channel *carries information the model lacks*, not whether a model could have learned a
 249 better representation from it. We retrain four arms at matched budget and data on two encoders,
 250 differing only in encoder input — and, for J1 against J0, in whether the surface branch exists at
 251 all, giving the design-valid arm *more* capacity: J0 backbone only; J1 adds point cloud, normals and
 252 geometry-derived hbond with hp zeroed (Route B); J2 design-valid predicted chemistry; J3 native
 253 chemistry (Route A). Two further pairs (B0/B1, G0/G1) test descriptor reach and a jointly trained
 254 geometry encoder (Appendix B).

255 Route A (J3 − J2) is +45.01 pp [+33.07, +60.43] released and +64.14 pp [+63.38, +64.92] on
 256 Simple2 — essentially the entire surface-branch gain on both. The released interval is wide because
 257 that arm is seed-unstable on *that* architecture while Simple2 spans 0.36 pp: direction robust,
 258 magnitude not. The design-valid arm J2 − J0 is −0.38 pp and −0.13 pp, so under a satisfiable
 259 contract the surface branch returns the model to backbone-only — deployment realism, not a route-B
 260 contrast, since it varies geometry and chemistry source at once.

261 Route B — four qualifying jointly trained contrasts in Figure 2b against the predeclared threshold.
 262 All four point estimates fall below +1 pp, while the *same* apparatus clears +1 pp when the channel
 263 really is $f(B)$. Only the two ten-seed contrasts are interpreted inferentially; their largest marginal
 264 95% upper endpoint is +0.22 pp. The two three-seed contrasts are directional checks without coverage
 265 claims (seed counts and controls: Appendix B, Table 7). Read together over seed means at a fixed
 266 chain set the worst of the four still sits at −0.01 pp, also as a direction check rather than a bound. The
 267 four contrasts and the positive control share one axis in Figure 2b.

268 The claim ceiling follows from the two calibrated contrasts, which bound Route B under weak tested
 269 instruments; the other two are consistent directional checks, and none supports an absence claim. The

instruments are §3.2’s small controlled models on Shrake–Rupley point clouds (Shrake & Rupley, 1973), not the solvent-excluded meshes the substrate methods use (Connolly, 1983; Sanner et al., 1996), at ≈ 23 points per residue against the pipeline’s 53.3. A denser surface or better mesh encoder may still realise Route B; our protocol would detect it. Route B in *surface* systems is open, not refuted.

A positive Route-B calibration appears in the last row of Figure 2b. A protocol that only ever returns nulls is hard to trust. Route B is a gain from $C = f(B)$ while the model *already has* B — $W_0 = M(B)$ against $W_1 = M(B, f(B))$, not a weak baseline against $f(B)$ alone. Over 10 seeds at matched split, budget and backbone stack, adding backbone-derived features carrying no sequence identity gives $W_1 - W_0 = +2.05$ pp [$+1.57, +2.53$], clearing $+1$ pp on its lower bound, and alignment beats a shuffled control by $+1.00$ pp [$+0.57, +1.45$]. The call is not automatic — the same append without the finite-head restriction did not clear $+1$ pp — so Component 6 is *sensitive* as well as specific (arm design: Appendix B). This calibration establishes sensitivity to a realizable Route-B effect; whether published surface meshes realize that route remains open.

5 Discussion

The failed hypotheses define the paper’s boundaries. The independence law $1 - \text{acc} = (1 - \text{base})(1 - \text{ceil})$ was REJECTED at 5.43 pp MAE against a 4 pp bar; an error-overlap detector was REJECTED after a ProteinMPNN threshold misclassified 6/6 ESM-IF1 backbone channels; and a coverage-only account was REJECTED at 3.50 pp against a 1.5 pp bar. A fourth prediction INVERTED, and we RETRACT the fold-quality family because two external judges disagreed in sign. All five remain to bound the contribution (Appendix C, Table 12).

Full-atom meshing adds only $+2.44$ pp decodability over backbone-only meshing, far below the channel’s 85.89% ceiling, so provenance remains live for geometry. The subject pipeline has no published checkpoint; §4.2 therefore uses a second system and does not run components 3 or 6 there. Joint instruments are small and supported by frozen full-strength models. External transfer includes a post-2022 bridge but a modest ESM-IF1/PiFold core. We make no fold-quality or prevalence claim; Appendix D states the full bounds.

Before attributing a derived channel’s gain to representation, apply three checks: (1) fixed-coordinate substitution (89.12% of values move versus 0.00% for geometry); (2) standalone capacity (85.89% Bayes recovery versus an 8.86% floor); and (3) marginal-decodability matching, where equal ceiling but unequal payoff points at provenance. Only the source intervention identifies provenance; the other checks are triage. Report all three.

By argument, surface-vs-no-surface ablations cannot identify the route (§1.3); by experiment, §§4.1–4.4 establish provenance and route bounds in the tested systems. Repairability supports mechanism rather than identification: fixed-weight neutralisation drops 6/6 checkpoints, layer-0 replay restores 73–100% of the gap, and short retraining restores parity (Appendix E). The protocol may transfer to other derived channels; the verdict need not. Appendix D maps four identification threats, including weak instruments and “a bug in one pipeline,” to controls.

6 Conclusion

Identification is a property of the comparison, not its magnitude: large gaps did not reveal their source. We do not estimate the checkpoint-free subject pipeline as shipped; a second released checkpoint transcribes a counterfactual code table. For AI in biology, audit derived inputs before attributing ablation gains to representation. Design-valid surface gains remain possible. Recurrence in other derived channels is open; our protocol makes it testable.

LLM-use disclosure

Large language models and agentic coding tools assisted literature triage, experiment planning, code and manuscript drafting, and artifact auditing. The authors verified every reported value against retained artifacts and remain responsible for the paper’s content.

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400

A Experimental details

A.1 Released-path arms for the subject pipeline

Four arms differ only in what the surface branch carries, at matched budget: A0 the released target-derived chemistry, A2 that channel set to a constant, A3 its values shuffled within the chain (marginal preserved, alignment destroyed), A5 no surface branch. They retrain the released encoder under our decoder; they are not published-checkpoint behaviour — none exists for that pipeline — and do not reproduce the subject paper’s recoveries.

Alignment, not magnitude, is what they show. $A2 - A3 = +0.32$ pp [$+0.12, +0.56$]: shuffling channel values within a chain makes the arm slightly *worse* than a constant, while the same values aligned to residues ($A0 - A2$) recover a large gap under this retrain ($+53.17$ pp, two-level bootstrap [$+44.60, +63.78$], $n=3$; the seed range itself is [$+44.48, +63.84$]). Same values, wrong residues, no payoff. Neutralising the channel leaves the surface branch slightly worse than having none ($A2 - A5 = -0.51$ pp [$-0.76, -0.27$]). All three are two-level bootstrap intervals at $n=3$ and are read as descriptive under the rule stated in Table 7’s caption — direction robust, magnitude not (per-arm recoveries: Table 5). §4.2 runs the same alignment control on a *released* checkpoint, where it costs -46.69 pp.

This appendix expands the population, benchmark and arm-level detail behind §§3.2 and 4.2; Appendix B carries the seed-level constituents.

Matched-decodability panel detail (Table 2). The external panel sizes are ProteinMPNN 68 chains / 9,540 residues (EXP38), with ESM-IF1 and PiFold on a shared 23-chain / 3,230-residue core (EXP39–EXP40) so the three families are read on identical structures; base recoveries there are 52.31%, 60.84% and 53.50%. Two of the three seeded 2-cluster target draws tied exactly on ceiling (17.126%); the alternative pays $+11.60$ pp on ProteinMPNN and was not present in the primary ESM-IF1 artifact set. PiFold was not run on the CATH panel, which is why that cell is empty.

Base-rate reconciliation (§3.2). On our 928-chain population frozen ProteinMPNN recovers 46.07% and frozen ESM-IF1 57.53%. The published figures are 52.4% on native backbones (Dauparas et al., 2022) and 51.6% on the CATH 4.3 “All” split (Hsu et al., 2022); PiFold reports 51.66% on CATH 4.2 and LM-Design 55.65% on the same split. The differences come from population and decoding rule, not from any reimplementations: we run the released weights unmodified, on a subset of the CATH test split, with our own decoding. Our ProteinMPNN number on TS500 is 52.80%; no published ProteinMPNN TS500 figure exists to calibrate it against. §3.2 states why none of this reaches a claim. We record the reconciliation here because leaving it out would be the kind of silence that makes a reader wonder what else was not checked.

Named populations (§3.2). The body states the rule; this table states its per-population consequences.

Table 3: Named populations (Table 3). P-FROZEN-FULL is a strict superset of P-FROZEN’s 197 test chains; the counts differ because the released pipeline evaluates on its own subset of the benchmark split. No chain was re-split, reassigned or reweighted. Applying §3.2’s rule keeps 928 of the 1,120 test chains (83% by chain, 64% by residue) and drops everything longer than 258 residues, which is what bounds the frozen-model and joint-training results to that window; TS50/TS500 carry no upper bound, so §4.6 does reach longer chains. P-FROZEN additionally requires at least 100 surface points per chain, caps each cloud at a seeded random 4,000 points, and takes a seeded 593/197 subsample. P-CEIL is unfiltered, which is why its frequency floor (8.86%) differs from the 8.85% used on the joint-training population.

Label	Definition
P-RELEASED	3,946 train / 790 test family-disjoint CATH 4.2 chains, 3 seeds
P-FROZEN	593 train / 197 test chains; 86,419 / 26,289 residues
P-FROZEN-FULL	928 test chains / 125,401 residues (83% of the CATH 4.2 test split by chain, 64% by residue)

Label	Definition
P-CEIL	195,631 test residues — analytic decodability bounds only
P-TS50 / P-TS500	50 / 500 chains from the subject paper’s other two benchmarks (O’Connell et al., 2018)

434 Benchmark transfer, per benchmark (§4.3). Codebook statistics and λ fitted on CATH, applied
435 unrefitted, before and after removing every chain that overlaps either CATH population. TS50 shares
436 3 exact sequences with CATH test and 1 with CATH train; TS500 shares 10 and 9.

Table 4: Benchmark transfer before and after decontamination. CATH values are Table 2’s and are not repeated; CATH is not decontaminated against itself. Removal uses an exact-identity criterion, not a similarity-clustered one, with Retained the surviving chain count. Both interval columns come from the same bootstrap run, so the before/after comparison is not confounded by resampling noise. Removing every overlapping chain shifts the contrasts by +0.02 and −0.02 pp.

Benchmark	n	Contrast	95% CI	kept	Contrast (decont.)	95% CI (decont.)
TS50	50	+12.05 pp	[+11.13, +12.96]	46	+12.08 pp	[+11.14, +13.05]
TS500	500	+12.20 pp	[+11.97, +12.43]	481	+12.18 pp	[+11.95, +12.41]

437 Released-encoder arm recoveries. The contrasts quoted in §4.2 are two-level bootstrap estimates
438 over seeds and chains, as §3.3 requires for trained arms, and differ slightly from differences of this
439 column.

Table 5: Released-encoder arms under matched training (P-RELEASED, 3 seeds). Arm labels are the released experiment’s own; A1 and A4 are internal variants that carry no claim here. Micro recovery is residue-weighted; the contrasts in the text are paired per-chain bootstrap estimates and differ slightly from differences of this column.

Arm	Channel	Micro recovery
A0	native hp	74.76% (sd 9.28)
A2	constant	21.50%
A3	shuffled within chain	21.18%
A5	surface branch off	22.07%

440 Worked decode of one chain. 1a1x.A, the first chain of the test split (108 residues), decoded from
441 the channel value alone with no model. Eight of the first ten residues are recovered exactly by table
442 lookup; the two exceptions fall in the single collision class.

Table 6: The codebook applied to one real chain, 1a1x.A. Chain-level, 80 of 108 residues (74.07%) are uniquely resolved, against the 78.65% population figure.

position	native residue	channel value	decodes to
1	G	−0.4	G uniquely
2	S	−0.8	S uniquely
3	A	+1.8	A uniquely
4	G	−0.4	G uniquely
5	E	−3.5	one of {D, E, N, Q}

position	native residue	channel value	decodes to
6	D	-3.5	one of {D, E, N, Q}
7	V	+4.2	V uniquely
8	G	-0.4	G uniquely
9	A	+1.8	A uniquely
10	P	-1.6	P uniquely

Populations. Table 3 defines them; this appendix adds only provenance. The two split files are hash-pinned — `chain_set.jsonl sha256 1944d21b...`, `chain_set_splits.json sha256 8e9a587a...` — and no chain was re-split, reassigned or reweighted by us; the selection rule that produces each evaluated population from those files is stated in Table 3.

Models. §3.2 names the four systems and their parameter counts. Nothing was modified in the two frozen models: published weights, published tokenisers, inference only.

Training. Every trained arm uses AdamW (`lr 1e-3`, `weight_decay 1e-4`) with a OneCycleLR schedule at `max_lr 1e-3`, gradient-norm clipping at 1.0, one chain sampled per step, and a budget of 3,000 steps. All trained arms use seeds 0, 1, 2. The controlled head is 128 hidden units with three message-passing rounds. *Matched budget* means the same optimiser, schedule, step count, clipping and seed set across the arms of a comparison; no arm received a hyperparameter search the others did not.

Compute accounting. The primary released-encoder and frozen-model work used rented A40 GPUs where the retained logs record the device; the SurfPro Stage-2 seed expansion used 24 GB RTX 4090 pods. Controlled joint-training arms also ran locally on Apple MPS, and analytic ceilings, audits and re-analyses were CPU-only. The recorded rented-GPU spend was \$9.68: \$1.93 through the first complete study freeze and \$7.75 for the prespecified extra-work round; all pods were stopped after outputs were downloaded and hash-verified. This is a cost ledger, not a hardware-normalised compute estimate; exact commands, update counts, seeds and environment versions are retained in the evidence manifests.

Uncertainty and injection. The two bootstrap procedures and the one-parameter injection rule are stated where they are used, in §3.3 and §4.3. The remaining details: the λ grid is given in §4.3 and contains 0, so a fitted $\lambda = 0$ leaves the base logits untouched and yields a gain of exactly zero rather than an estimate of one; the trained-MLP robustness variant is fitted with the same optimiser and budget as the arms it checks, and λ is refitted independently for every channel rather than carried across.

Reproducibility detail. Cold replay runs from nothing: it fetches and hash-verifies the dataset, clones the published model, recomputes logits and reproduces the central contrast — 3/3 objects, plus 12/12 headline values. Binding: every distinct decimal value in the claim contract occurs in a retained result file (0 unbound), and every decimal in the manuscript occurs in the contract, a control file or a retained artifact (0 unbound). Consistency: an automated audit enforces cross-file numeric agreement and bans retracted wording (0 FAIL / 0 WARN). Pinned inputs: dataset `chain_set.jsonl sha256 1944d21b...`; released encoder sources vendored in-project and hash-verified against the pinned commit. The certified package holds the manuscript in source and rendered form, all six figures with the script that regenerates them, the claim contract and evidence manifest, the three audit scripts and the two replay scripts, with a `sha256sum`-format checksum file; certification extracts it into a disposable directory, re-verifies every member, compiles every script and scans for credential-shaped strings and absolute local paths. The experiment runners are deliberately excluded: 25 of 103 embed paths from the machine they were written on, and the reproducibility claim rests on the replay scripts, which do not.

Reference verification. The bibliography is checked against two trails: the comparator papers against the project’s comparator registry, and sixteen methodological and dataset sources against Crossref — thirteen resolved by bibliographic query, three adjudicated against a named local source because conference proceedings and Nature’s AlphaFold record do not resolve cleanly. The audit trail is `Comparators/reference_verification.json`. The two trails do not cover every bibliography entry; items added after a trail was built carry their own source note instead.

B Per-seed results

The blind comparison that motivated this work found explicit natural-unit uncertainty in one paper of seven. Reporting an interval is not the same as letting a reader see the spread it came from, so the seed-level constituents of every trained contrast are given here, in Tables 8 and 9.

The four design-valid contrasts of §4.4, which Figure 2 plots against the predeclared threshold:

Table 7: The four qualifying design-valid contrasts under joint training on P-FROZEN (197 test chains), with two-level bootstrap intervals (Table 7). Both outside-J experiments also ran a shuffled-descriptor control (B2, G2); neither carries a claim. Rows 1, 2 and 4 use the declared 4,000-point construction; preliminary 1,200-point-cache runs are excluded. Row 3 does not consume the point cloud. Row 4 uses ten seeds: its preliminary three-seed estimate was -0.23 pp $[-0.45, -0.00]$, the only Route-B row whose descriptive interval excluded zero, whereas the ten-seed result is -0.00 pp $[-0.21, +0.22]$. The rule is applied symmetrically: three seeds cannot calibrate a tail, so every three-seed two-level interval in this paper is read for direction, not for coverage — including the three that exclude zero in our favour (J3 – J2 in §4.4, A2 – A3 and A2 – A5 in Appendix A). Across ten seeds the point estimate moved to zero and the interval widened — the upper bound rose from -0.00 to $+0.22$ pp, away from zero rather than toward it — so the family still clears $+1$ pp by a factor of four and rests on no zero-excluding Route-B row.

#	Test	Contrast	Δ re- covery (pp)	95% CI	n_seed
1	released encoder, geometry-only surface	J1 – J0	-0.22 pp	$[-0.52, +0.07]$	3
2	Simple2, geometry-only surface	J1 – J0	-0.08 pp	$[-0.24, +0.08]$	3
3	rich backbone descriptors fed straight to the head	B1 – B0	-0.09 pp	$[-0.34, +0.16]$	10
4	learned encoder over rich surface geometry	G1 – G0	-0.00 pp	$[-0.21, +0.22]$	10

Table 8: Per-seed trained contrasts, in percentage points. The first eight rows are the J-series on the two encoders; the last two are the design-valid pairs that sit outside it — `head only` is B0/B1, descriptors fed to the decoder head with no surface encoder at all, and `learned` is G0/G1, a jointly trained encoder over surface geometry — whose per-seed values are the arm means differenced seed by seed. The `n` column is how many seeds the row’s mean and sd are computed over; the three seed columns always show seeds 0-2, so for a ten-seed row they are a window rather than the whole row. `seed sd` is the population standard deviation over those `n` seeds, not the sample estimate — at three points the two differ by a factor of $\sqrt{3/2}$, so a reader recomputing the column should know which convention it uses. It describes spread across seeds and is not an inferential interval; the chain-level bootstrap intervals for the four design-valid contrasts are in Table 7. The released encoder’s J3 – J2 spans 32.73–60.50 pp across three seeds; that range, not any single seed, is what should be quoted. The design-valid contrast J1 – J0 is seed-stable (sd well under 0.1 pp) on both encoders, even though its two-level bootstrap interval on the released encoder now includes zero — seed-stability and interval width are different properties, and the interval’s width is dominated by chain-resampling noise, not seed variance.

encoder	contrast	seed 0	seed 1	seed 2	mean	seed sd	n
released	J1 – J0	–0.32	–0.15	–0.18	–0.22	0.08	3
released	J2 – J1	–0.11	–0.34	–0.05	–0.17	0.12	3
released	J3 – J2	+41.80	+32.73	+60.50	+45.01	11.57	3
released	J3 – J0	+41.37	+32.24	+60.27	+44.63	11.67	3
Simple2	J1 – J0	–0.09	–0.14	–0.00	–0.08	0.05	3
Simple2	J2 – J1	–0.04	–0.21	+0.11	–0.05	0.13	3
Simple2	J3 – J2	+64.21	+64.38	+63.81	+64.14	0.24	3
Simple2	J3 – J0	+64.08	+64.04	+63.92	+64.01	0.07	3
head only	B1 – B0	–0.11	–0.16	+0.08	–0.09	0.12	10
learned	G1 – G0	–0.12	–0.10	–0.35	–0.00	0.17	10

Table 9: Per-seed mechanism probes, the constituents of the ranges quoted in §4.5.

encoder	seed	A0 recovery	A2, channel removed	layer-0 restored
released	0	64.82%	7.7599%	80.79%
released	1	55.62%	7.7713%	89.97%
released	2	84.48%	6.7899%	73.20%
Simple2	0	86.20%	6.7823%	100.0000%
Simple2	1	86.47%	6.7861%	100.0000%
Simple2	2	86.15%	6.7823%	100.0000%

495 SurfPro matched retraining, per seed (§4.2). Nine arm/seed cells, 887 chains each, on SurfPro’s own
496 official scorer; the code table is the only thing that changes between arms, and data order and update
497 budget are paired by seed. Arm means are Table 10 and the contrasts they produce are Table 11.
498 The `seed sd` column of Table 10 is the *sample* standard deviation ($n - 1$), the convention of the
499 EXP-45 artifact these rows are quoted from — unlike Tables 8 and 9 above, which use the population
500 convention. At three seeds the two differ by $\sqrt{3/2}$, so the convention is stated per table rather than
501 assumed.

Table 10: Per-seed SurfPro Stage-2 arm means (Table 10), in % recovery. seed sd is the sample ($n - 1$) standard deviation, matching the artifact.

arm	seed 1	seed 2	seed 3	mean	seed sd
native (released table)	57.3335	57.1449	56.8405	57.1063	0.2487
scram	56.9508	56.9499	56.5934	56.8314	0.2061
scram2	57.4244	56.9926	57.1035	57.1735	0.2243
part18 (random partition)	60.9145	60.0491	60.3971	60.4536	0.4355

Table 11: SurfPro Stage-2 paired contrasts (Table 11). The paired treatment is primary (data order and budget matched by seed); the unpaired treatment is reported as the conservative alternative, and every semantics interval sits inside the predeclared 1 pp band in both directions under both. Read for direction, not coverage, under the same three-seed rule applied above. `scram - native` is the one semantics contrast whose paired interval excludes zero and whose sign holds at all three seeds: a real but small table-specific effect, which is why the paper says the specific assignment may matter at the few-tenths level and not that semantics are inert. `scram2 - scram` is sign-consistent at all three seeds but 3-of-3 carries p of about 0.25 and its paired interval includes zero, so it is reported as plausible, not established.

contrast	per seed (pp)	mean	paired 95% CI	unpaired 95% CI	% of the 47.82 pp gap
scram - native	-0.383, -0.195, -0.247	-0.275	[-0.516, -0.034]	[-0.707, +0.157]	0.575%
scram2 - native	+0.091, -0.152, +0.263	+0.067	[-0.451, +0.586]	[-0.366, +0.500]	0.141%
scram2 - scram	+0.474, +0.043, +0.510	+0.342	[-0.304, +0.988]	[-0.088, +0.772]	0.715%
part18 - native	+3.581, +2.904, +3.557	+3.347	[+2.393, +4.301]	[+2.780, +3.915]	7.00%

502 The eleven-arm capacity family. All 11 pre-specified arms are scored — zero, `part2`, `part3`, `part5`,
503 `part8`, `part12`, `native`, `scram`, `scram2`, `part18`, `ident`. Recovery is strongly rank-associated
504 with the channel’s standalone capacity (Spearman 0.9636 on the coverage scorer, 0.9909 on the
505 official one; grouped-chain interval lower bounds 0.9636 and 0.9727, both clearing the registered
506 0.90). No functional form survives: the pre-registered *linear* law failed (`part5` missed by +4.86
507 pp against a 3.7 pp tolerance, and the tolerance was not widened), and the fitted curvature changes
508 *sign* between the two scorers, so neither “linear” nor “concave” is scorer-invariant. Only the rank
509 association is claimed.

510 Reproducibility limit, stated plainly. The nine Stage-2 checkpoints (≈ 73 MB each) are on stopped
511 pod volumes and are not in the local package. Their sha256 values are recorded, so a recovered
512 checkpoint can be verified, but the scores, per-chain outputs, commands and logs are what is retained:
513 this arm is not part of the advertised cold replay, and we do not describe it as one.

514 The two tables come from different experiments and their recovery columns should not be compared
515 row by row: the first is the four-arm joint-training comparison on P-FROZEN, the second is the

fixed-weight pathway probe on checkpoints trained separately for it. Both use seeds 0-2, and both show a related but not identical pattern — the released encoder’s *recovery* is seed-unstable (spanning 32.73–60.50 pp on the J3 – J2 contrast and 55.62–84.48% here) while the *necessity* probe on it is not (0.98 pp across seeds); the *localisation* probe is intermediate — seed-variable (73.20–89.97%) but far less so than recovery itself, which is the released encoder’s version of the layer-0 remainder described in §4.5.

522

523 C Complete predeclared hypotheses

524 §5 states the five outcomes in prose. The ledger is here, followed by each hypothesis in full: what
525 was predicted, how it was tested, and what came back.

Table 12: Predeclared hypotheses that failed (Table 12). Each threshold or direction was fixed before the run that tested it; all are retained because they bound the contribution.

Hypothesis	Predeclared test	Outcome
Channel payoff follows an independence law $1 - \text{acc} = (1 - \text{base})(1 - \text{ceil})$	MAE < 4 pp	REJECTED, MAE 5.43 pp
Error-overlap lift gives a transferable provenance detector	threshold transfers across models	REJECTED — misclassifies 6/6 backbone channels on the second model
Payoff depends only on the <i>count</i> of labelled residues	structured vs random placement within 1.5 pp	REJECTED — structured placement deviates 3.50 pp
The channel improves designs, not just recovery	two external judges agree	RETRACTED — the two judges disagree in sign on the same designs; no fold-quality claim is made
Exposed residues gain most from a full-atom surface	predicted direction	INVERTED — the gain concentrates where the backbone surface is <i>sparsest</i> (17.69% vs 11.61%)

1. An independence law relating base rate and channel ceiling. Predicted that a channel’s payoff satisfies $1 - \text{acc} = (1 - \text{base})(1 - \text{ceil})$, i.e. that knowing a model’s base recovery and a channel’s standalone ceiling determines the gain. Predeclared acceptance: mean absolute error below 4 pp across the codebook sweep. Observed 5.43 pp. Rejected, and it stays rejected: a later relation of ours ($\text{gain}(1) = (1 - \text{base}) \times \text{ceil}$) is this same law at full coverage and was withdrawn for that reason rather than developed.
2. Error-overlap lift as a transferable provenance detector. Predicted that the ratio of a channel’s error overlap with the base model separates target-derived from backbone-derived channels, with a threshold that carries across models. On ProteinMPNN the two families are disjoint — target-derived [1.02, 1.49], backbone-derived [1.57, 1.61] — which is why it looked like a detector. On ESM-IF1 they overlap: target [1.00, 1.33], backbone [1.35, 1.38]. Any threshold fitted on the first model misclassifies 6/6 backbone channels on the second. Rejected as a detector; retained only as a mechanistic observation about redundancy.
3. Payoff depends only on how many residues are labelled. Predicted that $\text{gain}(f) = f \times \text{gain}(1)$ holds for any placement of labels, not just random placement. Predeclared tolerance: structured placements within 1.5 pp of the random-coverage line. Observed maximum deviation 3.50 pp. Rejected, and the surviving relation now carries an explicit random-coverage qualifier wherever it appears.
4. The channel improves designs, not merely recovery. Predicted that two independent structural judges would agree that channel-conditioned designs are better. They disagreed: normalised excess of -0.0019 on one judge against $+0.3666$ on the other, over the same

547 designs. The disagreement has a mechanical explanation — the first judge saturates near
548 70% identity (design 2.4901 against control 2.5320) — but the point is that we had no basis
549 to pick the favourable judge. Retracted, and the entire fold-quality claim family was deleted
550 rather than re-tested with the judge that agreed with us.

551 5. Exposed residues gain most from a full-atom surface. Predicted direction: side chains reach
552 the surface only on exposed residues, so the identity information a full-atom mesh adds
553 should concentrate there. Inverted. The gain concentrates where the *backbone surface is*
554 *sparsest* — +6.08 pp there (17.69% against 11.61%) versus +0.53 pp where it is densest.
555 The stratifier that matters is local surface density, not burial. A related exposure-gradient
556 prediction failed in sign the same way (exposed +2.50 / +3.50 pp, buried −2.58 / −3.50 pp)
557 and that claim family was withdrawn.

558 Two of these — 1 and 3 — are the numerically comparable pair plotted in Figure 4. Hypotheses
559 4 and 5 failed in kind rather than in magnitude and cannot be placed on a ratio axis; 2 failed as a
560 *transfer* claim rather than against a numeric threshold.

561

562 **D Threats to identification and claim boundaries**

563 **D.1 Identification threats and corresponding controls**

564 These are the identification threats named in §5; each control is stated only to the evidentiary ceiling
565 it supports.

566 “Controlled models are too weak for Route B.” Conceded, at the ceiling stated after Figure 2: the
567 frozen panel bounds *information addition* at published scale, the joint panel bounds *representation* at
568 ours, and neither bounds representation at published scale. That is why §4.4 records Route B as open
569 rather than refuted.

570 “Marginal is not conditional decodability.” Deliberate. Matching the conditional quantity would erase
571 the very route difference the design isolates; equal marginal content with a ≈ 12 pp payoff gap is the
572 point of §4.3, not an oversight in it.

573 “ $\lambda = 0$ hides a nonlinear benefit.” Tested rather than argued: the trained combiner (a 256-256 MLP, not
574 a linear blend) returns negative backbone gains on both the matched-ceiling and the learned-predictor
575 codes (§4.3), so the null does not depend on linearity.

576 “This is a bug in one pipeline, not a contribution.” A scope objection, not an evidential one. The
577 product is the estimand (§1.3) and the protocol (§3.1), and the same reading applied to MapDiff
578 returns *clean by construction* (§2) — a protocol that only ever accused would not be worth having.

579 Input-capacity boundary. Full-atom meshing buys +2.44 pp decodability over backbone-only meshing
580 across 100 chains, far below the chemistry channel’s 85.89% ceiling: the geometry channel is a
581 capacity boundary, not a second provenance finding.

582 **D.2 Released-checkpoint error fingerprint**

583 The 20x2 code table of §4.2 resolves the 20 residues into 18 classes, colliding exactly {D,E} and
584 {N,Q}. Over 137,346 aligned residue pairs, the released checkpoint’s four largest confusions are
585 exactly the four ordered pairs inside those two classes — Q → N 32.58%, D → E 32.34%, E →
586 D 21.43%, N → Q 17.86% — and the fifth largest, C → L, is 2.4 times smaller at 7.53%. The
587 control that separates “reads the table” from “knows chemistry” is I/L/V: as similar biochemically
588 as D/E, but given three distinct values by the table, and separated by the model (I → L 7.21%). The
589 alignment used here charges an explicit mismatch and gap cost and is used only to read confusions,
590 never to compute a recovery number.

591 **D.3 Claim boundaries**

592 These eleven boundaries define the scope of the paper’s claims:

- the paper contributes a causal diagnostic protocol, not a new inverse-folding model;
- $I(Y; f(B) \mid B) = 0$ is motivation, not novelty, and not a theorem;
- Route B is 4 qualifying joint-training contrasts; the 2 frozen-model falsifiers are reported separately;
- say “exclude benefits of +1 pp or larger under tested conditions”; never write “Route B is absent”;
- say 3 named benchmarks, 1 primary data authority, 0 independent external-data replications;
- say benchmark transfer, not independent-data generalisation;
- do not generalise from the tested label-free arm to all molecular surfaces;
- do not claim behaviour of unavailable published checkpoints;
- do not claim fold quality, biological function, field prevalence, intent or misconduct;
- full-atom meshing is an input-capacity boundary (capacity gain stated in §5), not checkpoint use;
- random-coverage and fixed-codebook relations are secondary.

607

608 The subject pipeline’s own channel ablation. The released paper states its surface features are
 609 “computed from local atomic geometry and element types, not from residue identity labels” and that
 610 the pipeline does “not use: (i) residue identities ... during the surface generation pipeline”. Its
 611 experimental appendix takes three channels from MaSIF — Poisson–Boltzmann electrostatics via
 612 APBS, hydrophobicity, and free electrons/protons — and reports: “*After a further ablation study, we*
 613 *discover that the hydrophobicity and the charge are pivot to the performance improvement while the*
 614 *electrostatics is not necessary.*”

615 Of those three, the two named as pivotal are the two computed by table lookup on the residue name,
 616 and the one named as unnecessary is the one solved from atomic geometry. §4.1 measures the first of
 617 them moving 89.12% under fixed-coordinate permutation against 0.00% for the geometry channel.
 618 We report the discrepancy between a stated scope and a released implementation, and make no claim
 619 about how it arose; a removal ablation of the kind quoted is exactly the comparison §1.3 shows cannot
 620 separate the two routes, so it does not settle the question either way.

621 Two nearby audits, and what they do not supply. An RNA audit framework quantifies which residue-
 622 level annotations a coarse geometry representation supports, isolating an information source with an
 623 ablated seven-atom versus six-atom control (Gow et al., 2026) — the same audit spirit, on backbone
 624 geometry rather than a derived chemistry channel, with no determinism claim and no code table to
 625 permute. A separate enumeration searches subsets of a ten-letter amino-acid pool for de novo design
 626 (Pubal et al., 2026); our enumeration is over partitions of the full twenty-letter alphabet and scores a
 627 released model’s codebook, which is a different object and a different question.

628 The matched ceiling is a frontier over our constructions, and we tested where it stops. §4.3 matches
 629 the two families at a $\approx 17\%$ marginal ceiling, and the natural objection is that this is far below the
 630 released channel’s 85.89%. Across the retained 48-pair sweep no backbone-derived channel reaches
 631 a train-estimated marginal ceiling above 21.50% on either frozen model, attained by `bbkmeans1024`.
 632 Two objections to reading that as a frontier are real, and we ran the test rather than answering them in
 633 prose.

634 *First, the estimator.* That 21.50% fits the per-cluster argmax on the same train residues it scores, and
 635 with ≈ 84 points per cluster the optimism grows with k . Refitting on train and scoring on held out
 636 test moves `bbkmeans1024` from 21.50% to 18.60% — a +2.95 pp bias. On target-derived codebooks
 637 of at most 18 classes the same bias is +0.14 to +1.37 pp, which *falsified* a predeclared prediction that
 638 it would stay under 1 pp: the target side carries a small optimism too, about half a point larger than
 639 we predicted, and an order of magnitude smaller than the backbone side’s.

640 *Second, the stopping point.* $k = 1024$ was where the original sweep stopped, not a limit. Pushing the
 641 same recipe to 4,096, 16,384 and 65,536 clusters makes the train-estimated ceiling keep climbing —
 642 25.73%, 36.51%, 74.05% — while the held-out ceiling turns over and collapses: 18.60%, 17.02%,
 643 14.39%, 12.58%. At 1.3 train points per cluster the train estimate is optimistic by 61.47 pp: it was

644 measuring memorisation, not capacity. Five clustering seeds at $k = 1024$ and 4,096 and three at
645 16,384 put every held-out spread below 0.41 pp.

646 So the honest frontier over our constructions is 18.60% held out, and it is saturated in the sense that
647 spending more clusters now *lowers* it. Two independent constructions land in the same region: the
648 learned residue predictor over 19 backbone-only geometric features reaches 21.43% train-estimated,
649 and full-atom meshing buys only +2.44 pp of decodability over backbone-only meshing. A three-class
650 recoding of the native residue already exceeds the whole backbone family at 23.79%.

651 What this does and does not license. The matched contrast sits at $\approx 17\%$ because that is the top of the
652 range in which *we* could build a design-valid channel. This is a bound over one k-means descriptor
653 family on this population, and no amount of saturating it can prove that no design-valid learned
654 representation exceeds it — a better one would raise the matched point, and §4.4 says so. The claim
655 is that the 85.89% contrast is un-run for want of a construction, never that it is impossible.

656 Capacity buys recovery only when the capacity is about the target. The same operation — more
657 classes — has opposite signs in the two families (Table 13).

Table 13: The ladder (Table 13), in pp of native sequence recovery on frozen ProteinMPNN. Target-derived payoff is strongly rank-associated with ceiling (Spearman +0.9779; +0.9448 on ESM-IF1); backbone-derived payoff is rank-associated *negatively* (−0.7638; −0.9411). The registered test was Spearman > 0.90 and it passed, but the relation is not literally monotone: sorted by ceiling, target payoff *decreases* at 7 of 26 adjacent ProteinMPNN steps and 6 of 17 on ESM-IF1, the largest single drop being −7.78 pp. Raising a design-valid codebook from the smallest to the largest buys 10.47 pp of ceiling and costs 0.92 pp of recovery. Inside the shared ceiling band no target-derived channel pays less than any backbone-derived one: the separation is +11.54 pp on ProteinMPNN and +9.75 pp on ESM-IF1. Point estimates over the sweep, not a distributional claim; all four predictions here were predeclared and confirmed.

family	channel	ceiling	Δ
backbone	burial3	11.03%	+0.00
backbone	bbkmeans4	13.71%	+0.00
backbone	bbkmeans16	17.00%	+0.00
backbone	bbkmeans64	18.41%	+0.00
backbone	bbkmeans256	19.73%	−0.16
backbone	bbkmeans1024	21.50%	−0.92
target	rand_k2_s0	16.66%	+11.54
target	rand_k3_s0	23.79%	+19.32
target	rand_k4_s0	29.30%	+24.89
target	rand_k8_s0	48.32%	+34.57
target	rand_k16_s1	71.94%	+44.51
target	kd+charge	90.45%	+48.30

658 The provenance shortfall. Table 2 pairs one channel from each family at a matched ceiling, and
659 a reviewer can fairly ask whether that single pair generalises. Fit the capacity–payoff relation on
660 target-derived channels only, read it at each backbone channel’s own ceiling, and report the gap.
661 Channels whose ceiling falls outside the target range are excluded rather than extrapolated.

662 This is an estimator, and we say which one. Our first version justified the interpolation by calling
663 the target relation monotone; it is not (Table 13), so that justification is withdrawn — piecewise-
664 linear interpolation *is* a local functional-form assumption, and this paper has already had one such
665 assumption rejected (§5’s independence law). The shortfall is therefore re-run under four estimators
666 fixed in advance and reported under all of them: linear interpolation, isotonic regression, cardinality-
667 averaged local linear, and the monotone lower envelope, the smallest prediction any non-decreasing
668 reading of the target curve permits. We take the *minimum*, not the largest.

Two further repairs. The original leave-one-out left out one row, but the random recodings come in three seeds per cardinality, so a held-out row was usually predicted from two near-duplicate siblings; the control is now a group holdout removing an entire cardinality at once (2.26 pp ProteinMPNN, 3.33 pp ESM-IF1, against the same bar). And the estimate now carries an interval: a joint chain bootstrap resamples train chains (moving every ceiling) and test chains (moving every gain) in the same replicate and reruns the whole estimator inside it, $B = 1000$.

So qualified, the estimator predicts its own family well and the backbone family very badly, always in the same direction:

Table 14: The provenance shortfall (Table 14), in pp of native sequence recovery, ProteinMPNN / ESM-IF1. Predictions come from the target-derived relation alone; the design-valid channels were never used to fit it. The shortfall column is the minimum over four predeclared estimators (linear interpolation, isotonic regression, cardinality-averaged local linear, monotone lower envelope), not the largest; the last column is the worst 2.5th percentile over all four under a joint chain bootstrap ($B = 1000$, train and test chains resampled together, the whole estimator rerun inside each replicate). The linear estimator alone gives +11.95 / +13.50 / +15.09 / +17.76 and +10.23 / +11.67 / +12.94 / +15.26. The shortfall grows with capacity on both models, and the smallest lower bound anywhere in the table, +9.65 pp, is still three times the grouped-holdout error of 2.26 / 3.33 pp inside the family the relation was fitted on. Two predeclared glosses failed here and are corrected rather than dropped: “an order of magnitude” (measured $5.1\times$ to $7.5\times$), and “monotone”. The `bbkmeans1024` bootstrap mean sits *above* its own point estimate because resampling train chains inflates a 1024-cluster codebook’s train-estimated ceiling — the same optimism measured at +2.95 pp above — so the lower bound, not the mean, is the operative number. A descriptive interpolation over a fixed sweep on two frozen models, not a law.

channel	ceiling	measured	min shortfall	bootstrap 2.5th pct
<code>bbkmeans16</code>	17.00%	+0.00 / +0.00	+11.54 / +10.04	+11.12 / +9.65
<code>bbkmeans64</code>	18.41%	+0.00 / -0.16	+12.11 / +10.46	+11.68 / +10.11
<code>bbkmeans256</code>	19.73%	-0.16 / -0.17	+12.28 / +10.48	+11.84 / +10.14
<code>bbkmeans1024</code>	21.50%	-0.92 / -0.83	+13.03 / +11.14	+12.71 / +10.94

The channel’s ceiling is not a property of the chemistry. §4.1’s 85.89% ceiling is the Bayes-optimal recovery of one particular partition of 20 residues into 17 classes — the one the Kyte–Doolittle scale happens to induce. There are 741,285 such partitions ($= S(20,17)$), each with a closed-form ceiling $\sum_b \max_{y \in b} p(y)$, so whether this codebook is unusually leaky is decidable by enumeration rather than by argument. We enumerated all of them on P-CEIL from residue frequencies alone: no model, no coordinates, no sampling. 82.04% have a strictly higher ceiling. The biochemical table ranks 608,152nd, at the 17.96th percentile; the best attainable reaches 95.12%, the median 88.50%, the worst 77.99%. A table drawn at random therefore leaks 2.61 pp more than the released one at the median.

The mechanism is arithmetic, not chemistry. Merging a block costs everything but its most common member, so a merge is cheap exactly when the mass it destroys is small. Kyte–Doolittle merges {D, E, N, Q} — the residues sharing hydropathy -3.5 — which carry 21.35 pp of mass and keep only E (7.24 pp), destroying 14.11 pp. The cheapest table merges {C, K, M, W}: the three rarest residues together with a common one, K (6.12 pp), which becomes the class’s prediction and so costs nothing — 4.88 pp in total. Chemical similarity and informational cheapness are unrelated, so a codebook designed on chemistry merges whatever is chemically alike, and chemically alike residues are not rare.

Controls fixed before the run reproduce §4.1 exactly — the 8.86% floor, and the Kyte–Doolittle ceiling on all three splits (85.89% test, 86.39% train, 86.44% validation), each against a separately stored value — and the ranking holds on all three (17.96th, 20.46th, 22.41st percentile). A predeclared

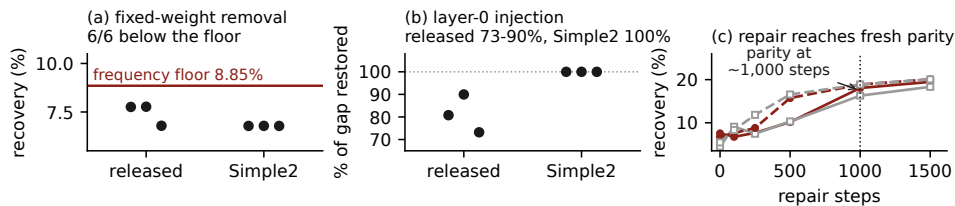


Figure 3: The mechanism chain. (a) With weights held fixed, neutralising the channel drops all six checkpoints below the frequency floor. (b) Layer-0 injection restores 73-90% of the gap on the released encoder and 100% on Simple2, so the commitment is made at layer 0 in both, completely in one. (c) Repair reaches parity with fresh training in about 1,000 steps; dark = repair, light = fresh, solid = released, dashed = Simple2.

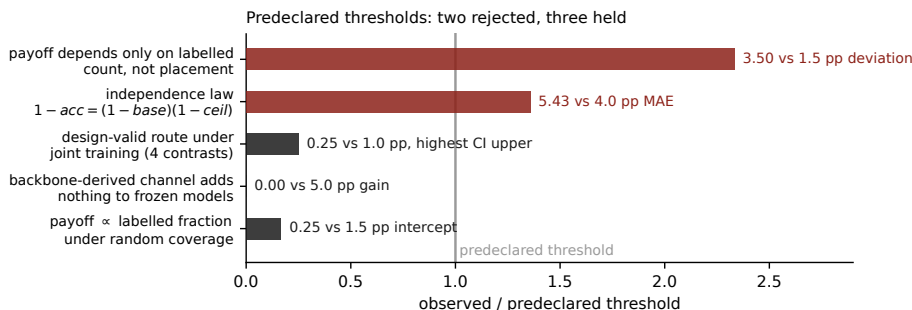


Figure 4: How the predeclarations actually performed. Five predictions whose thresholds were fixed before the runs that tested them, plotted as observed over threshold with the decision line at 1.0. Two were rejected. Three further predictions are not ratio-comparable and are listed in Table 6, so the five plotted here and the five in that table are different sets that overlap in two rows.

697 prediction failed: we expected the best table to merge only residues from the five rarest, and it merges
698 K, the sixth *most* common. The optimum is not “collide the rarest” but “collide the rarest with a
699 common one”. It is retained, like the five in §5.

700 This bounds a parameter-free lookup, not a trained model, and it is a statement about table design, not
701 about anyone’s intent: the Kyte–Doolittle scale is a hydropathy measurement and was never chosen
702 to maximise residue recovery. What it settles is that the provenance problem here is structural — it
703 follows from partitioning residue identity into 17 classes at all, not from which partition was chosen
704 — which is why §5’s burden asks for a capacity bound rather than for a defence of the scale.

705 E Additional figures and figure provenance

706 Table 7 carries the four qualifying design-valid contrasts of §4.4 with their intervals and per-seed
707 constituents; the body plots them against the predeclared +1 pp threshold alongside the positive
708 control, because a bound is only readable next to the calibration showing the same instrument can
709 clear it.

710 The complete mechanism chain of §4.5 has three stages. *Necessity*: with weights fixed, neutralising
711 the channel at inference drops recovery below the 8.85% frequency floor of P-FROZEN in 6/6
712 checkpoints. *Localisation*: injecting the native run’s layer-0 features into a channel-free forward
713 pass restores 73–90% of the gap on the released encoder and 100% (3/3 seeds) on Simple2 —
714 predominantly first-layer, not exclusively so. *Repair*: from below the floor, about 1,000 steps reach
715 parity with fresh training. The three stages are plotted together in Figure 3.

716 Figure 4 is placed here rather than in the body because Table 12 is the falsification record and this
717 is a secondary view of a different five-item subset: the three predeclarations that held, and the two
718 failures that are ratio-comparable with them.

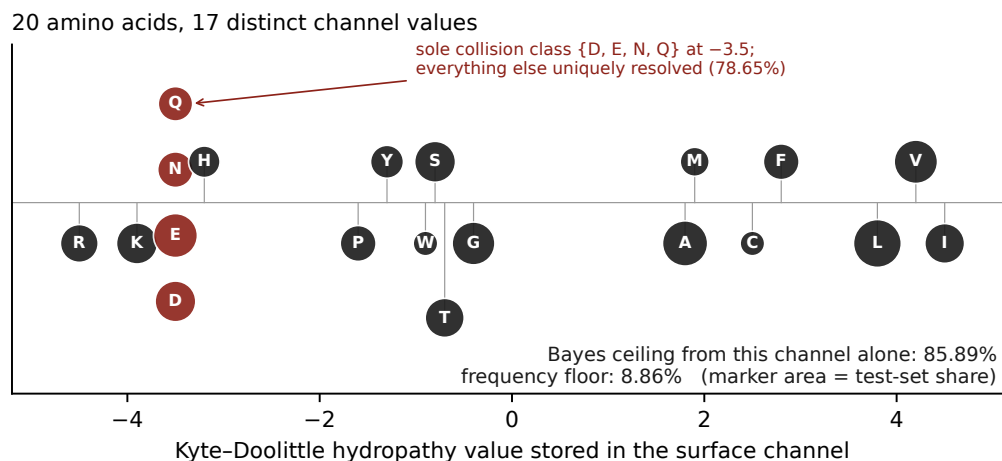


Figure 5: The artifact statistic. All 20 amino acids placed on the hydropathy axis used by the released channel; marker area is the residue's share of the test population. Twenty amino acids map to seventeen distinct values, with a single collision class, which is what puts the Bayes ceiling of the channel alone at 85.89% against an 8.86% frequency floor.

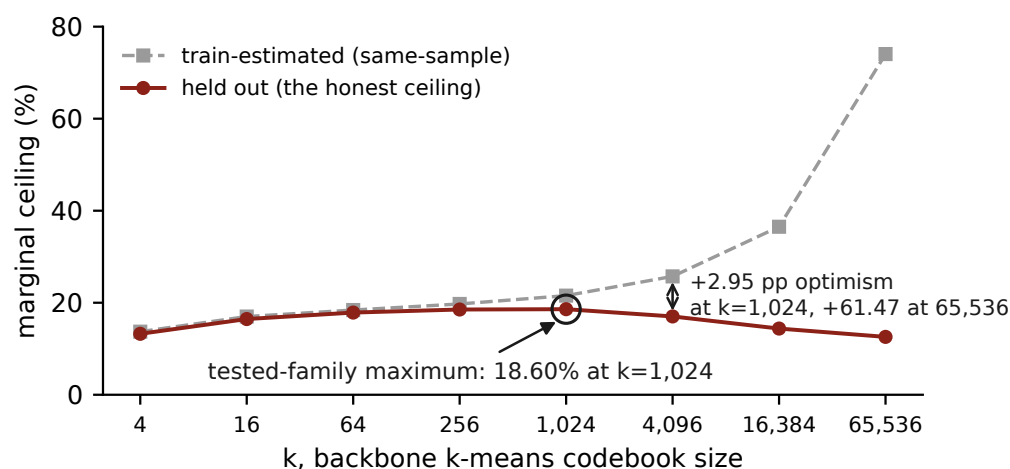


Figure 6: The frontier was an estimator artifact. Marginal ceiling of a backbone k-means codebook against k, estimated two ways. The train-estimated curve (the estimator EXP-47 used, which fits the per-cluster argmax on the residues it scores) climbs without limit to 74.05%. The held-out curve peaks at 18.60% at k=1,024 and then collapses: past that point extra clusters buy memorisation, not capacity. Shaded band is the observed range over clustering seeds.

719 The codebook panel illustrates the finding stated in §4.1 (17 channel values, sole collision {D, E, N, Q},
720 85.89% against 8.86%), so it is explanatory rather than load-bearing. The released channel's codebook
721 is plotted in Figure 5.

722 Every figure is captioned where it appears; this appendix records only how each was produced, which
723 the captions do not say.

724 EXP-49's re-estimation of the design-valid frontier is the one place where an estimator and the
725 quantity it estimates part company visibly, so the comparison of train-estimated against held-out
726 backbone ceilings is Figure 6.

727 All six are generated by one command, `make_figures.py`. A separate panel for the matched-
728 decodability contrast is omitted because Table 2 carries the same result with strictly more information.

Five are built directly from the retained artifacts and the script prints every plotted value at render time, so a figure cannot drift from the number it illustrates; the frequency floor and Bayes ceiling drawn as reference lines are read from `EXP1_hp_decodability/exp1_results.json` rather than typed in. One figure (`fig1_underidentification`) carries no measured value and reports that when it runs.

Table 15: The six rendered figures (Table 15), one schematic and five derived from retained artifacts. The build refuses to render a figure whose source artifact is absent, and the archive self-audit checks this list against the files it ships.

#	file	kind
1	<code>fig1_underidentification</code>	schematic
2	<code>fig4_route_b_bound</code>	artifact-derived
3	<code>fig5_mechanism_chain</code>	artifact-derived
4	<code>fig6_predeclared</code>	artifact-derived
5	<code>fig7_codebook</code>	artifact-derived
6	<code>fig8_frontier_saturation</code>	artifact-derived

The build refuses to proceed if a hand-written “Figure N” label disagrees with document order, if a figure anchor falls mid-paragraph, or if this table and the placed figure set disagree in either direction.

F Constructing the two routes

§1.3 argues that Δ_{ablation} cannot separate the routes. That argument is an identification claim, so it can be *built* rather than only asserted, and a construction is only worth reporting if its Route-A channel genuinely depends on the target — otherwise it proves equivalence by collapsing Route A into Route B.

A finite learner may condition on at most 64 cells. The backbone takes 4,096 states, so the learner cannot afford the raw backbone and must coarsen it. Two channels of identical cardinality (8) are offered to it:

- Route B — $C = g(B)$, a partition of the backbone states built by clustering them on the *population* conditional $P(Y|B)$, which is what a method designer legitimately does with training data. It never reads an instance’s Y .
- Route A — $C = a(Y, \text{noise})$, which names Y ’s class with probability p and is otherwise independent noise. p is calibrated on a disjoint sample so the match is never fitted to the evaluation set.

Over five seeds at 200,000 evaluation points each, with the same learner, baseline and estimator throughout:

Table 16: The two constructed pipelines of Appendix F (Table 16). Their ablation gaps agree to 0.07 pp across five seeds and their provenance does not: under an exact $\text{do}(Y)$ intervention the Route-A channel moves 31.25% of its values and the Route-B channel moves none. This is §1.3’s argument built rather than argued.

quantity	Route A	Route B
ablation gap Δ	11.65 pp	11.67 pp
largest per-seed gap difference	0.07 pp	0.07 pp
95% CI of the gap difference	[−0.06, +0.01]	[−0.06, +0.01]
channel moved under exact $\text{do}(Y)$	31.25%	0.00%

quantity	Route A	Route B
conditional information on the target	0.96 bits	0.00 bits

The ablation gap is blind and the intervention is not. Three properties make the construction non-degenerate rather than rhetorical. The Route-A channel carries 0.96 bits about the target given the *full* backbone state, so it is not a disguised function of B. A positive control — the same Route A with a higher leak rate — separates the two gaps by 9.72 pp, so the matching is a property of the construction and not of a blunt estimator. And the intervention recovers the true leak fraction to a mean absolute error of 0.0036, which is the same instrument reading 90.12% on the released system in §4.2.

The construction is deterministic: re-running it reproduces the result file bit-for-bit.

G Evidence governance

The two audits, and what each certifies. The looser one asks whether a manuscript decimal exists anywhere in the evidence base; it reports 0 unbound, and §6 claims only that. The stricter one asks whether a value sits *in* the artifact its own contract row names. Of 828 contract decimals, 743 bind there directly — exactly, rounded, or as a truncated quote of a longer stored decimal — and 70 are reproduced from that same file by arithmetic: a pairwise difference in the artifact’s unit or in percentage points, a sum, an interval midpoint, or a ratio. The remaining 15 are declared non-measurements, listed one at a time with a reason in the contract’s own section NC: a published constant (the Kyte–Doolittle value of the collision class), a dataset version, a predeclared threshold, and figures a row itself labels superseded. 0 are mis-scoped, and the check exits non-zero if one is.

An earlier adjudicator pattern matched only an ASCII hyphen, so every negative contract value was read as positive and charged against an artifact storing its negation. That inflated the miss count and hid five correctly scoped rows among the residue. Accepting the typographic minus this paper actually uses moved 134 values from miss to bound; the check was then mutation-tested — a falsified interval bound, a falsified point estimate and a falsified ceiling are each still caught.

Why the stricter audit is stated separately. A guard that verifies a number exists *somewhere* in a large evidence base verifies very little: the pooled set of artifact decimals here covers essentially the whole two-decimal grid over [0,100), so membership is nearly free. Row scoping restores discrimination by cutting the search space from that pool to the few dozen values in one named file. The declaration list is the escape hatch, and its size is printed on every run so it cannot grow unnoticed.

Entry point. README.md -> checks/archive_self_audit.py re-hashes every member, rebinds every decimal and confirms the headline values still render using nothing outside the archive. The $\approx$ 100 historical runners are not part of that advertised replay, and we do not claim they are.

H Additional analyses and disclosures

H.1 Transfer beyond CATH

Table 2 also tests transfer beyond CATH. Its external column is fitted on CATH and applied unfitted, and the input-side checks return CATH’s verdict on the same authority — a mean substitution move of 88.97% against 89.12% — so the provenance signature survives the move off CATH; codebook and λ transfer unfitted to TS50 and TS500 at +12.08 and +12.18 pp, exact overlaps removed (Appendix A, Table 4).

H.2 Additional related work

Partial-input baselines. Hypothesis-only NLI classifiers (Gururangan et al., 2018; Poliak et al., 2018) and benchmark-artifact work (Torralba & Efron, 2011; Geirhos et al., 2020) diagnose *datasets*, and are themselves one-directional (Feng et al., 2019); we diagnose a *derived input channel inside a method*, at matched marginal decodability (§4.3).

797 Ablation as an estimand. That a with/without contrast is algorithm-specific, and that its population
798 analogue vanishes whenever the feature is conditionally redundant given the rest, is settled in the
799 variable-importance literature (Williamson et al., 2023; Watson & Wright, 2021); a concurrent
800 position paper makes the parallel argument for interpretability (Lin & Liu, 2026). Route B is then
801 the creation of *usable* information by computation (Xu et al., 2020), which $I(Y; f(B) \mid B) = 0$
802 permits because Shannon information is unchanged. What none of this supplies is provenance —
803 whether the channel was computed from the target — which is the free variable when the method
804 builds its own input, and the one our protocol sets. Leakage taxonomies (Kaufman et al., 2011;
805 Kapoor & Narayanan, 2023) name illegitimate feature construction at study-design scale, not per-
806 example design-time legitimacy. Protein-ML work on split leakage and evaluator validity (Bushuiev
807 et al., 2024; Bernett et al., 2024; Surana et al., 2024; Korbeld et al., 2026; Wei et al., 2026) questions
808 evaluation with inputs fixed; we do the reverse. Nearest in spirit is a concurrent RNA audit of
809 residue-level information in tertiary geometry (Gow et al., 2026) and a concurrent enumeration of
810 amino-acid *subsets* for de novo design (Pubal et al., 2026); ours enumerates *partitions* of a shipped
811 codebook. Closest in framing (Dillon et al., 2026): one score, several information routes, models
812 grouped by what is available at scoring time — but that work decomposes variance by *controlling*
813 observed confounders where we intervene, and its grouping presumes one already knows what each
814 model reads, which is the assumption §4.1 breaks.

815 Surface-conditioned design. Surface fingerprints and their use in interaction design (Gainza et al.,
816 2019, 2023; Sverrisson et al., 2021) and surface-conditioned designers (Song et al., 2024; Li et al.,
817 2026; Wu et al., 2026) are the substrate; backbone-only inverse folders (Dauparas et al., 2022; Hsu et
818 al., 2022; Gao et al., 2023; Zheng et al., 2023) are bridges, and their ablations are the underidentified
819 comparison of §1.3.

820 H.3 Internal mechanism: necessity, localisation, repair

821 Component 7 opens the trained model and finds the commitment *necessary*, *localised* and *reversible*:
822 at fixed weights the channel cannot be removed at inference without dropping 6/6 checkpoints under
823 the frequency prior, replaying its layer-0 activations recovers most of that loss — predominantly at that
824 layer, not exclusively — and a short retrain restores parity. It *suppresses* the backbone pathway while
825 training rather than preventing one from being learned, which is what makes the defect repairable.
826 Values and Figure 3: Appendix E.

827 H.4 Reproducibility, disclosure and ethics

828 Reproducibility. Every number is bound to a retained artifact under a frozen claim contract,
829 checked mechanically, and a cold-start replay reproduces the central contrast from nothing: ev-
830 ery manuscript decimal is claim-table-, control- or artifact-bound (0 unbound), and under the
831 stricter row-scoped audit 0 of 828 contract decimals are mis-scoped. Manuscript, figures, con-
832 tract, audits and replay scripts are bundled as a hash-listed anonymous supplementary archive
833 (surfdesign_causal_decomposition_package.tar.gz). No reviewer-availability claim is
834 made until a venue-provided upload or anonymous link is verified. Mechanics, entry point and
835 what the replay excludes — including §4.2’s matched-retraining checkpoints, hash-bound but not
836 retained locally: Appendix G. §4.2’s arms use the subject system’s own released weights, data and
837 scorer, both hash-verified.

838 No claim about intent: everything we report is a property of code and measured behaviour. A
839 channel that reads the native sequence violates the input contract inverse folding is evaluated under
840 — a statement about the contract, not anyone’s intent, and we allege no misconduct. Disclosure
841 before publication: the diagnosis, external transfer and repair path are packaged for private notice
842 (Paper/DISCLOSURE_PACKAGE.md); the package is complete and held for transmission, and we do
843 not claim the email has been sent until that step is logged.